

H.S. 2nd Year — Computer Science and Applications (2026)

Solved Suggestive Question Paper — Code: /507

Q. No. 1 — Answer any eight ($1 \times 8 = 8$) [All parts solved]

(a) What is data hiding ?

Ans: Data hiding is an OOP concept in which the internal data (data members) of a class are hidden from outside access by declaring them as 'private', so that they can be accessed only through the public member functions of the class. It protects data from accidental or unauthorised modification and is a key part of encapsulation.

(b) What is abstract class ?

Ans: An abstract class is a class that cannot be instantiated on its own (no object can be created directly from it) because it contains at least one pure virtual function. It is designed to be used only as a base class from which other classes are derived.

(c) What is data abstraction ?

Ans: Data abstraction is an OOP concept of showing only the essential/relevant details (interface) to the user while hiding the internal implementation details. For example, a user of a class can use its member functions without knowing how they are implemented internally.

(d) What is inheritance ?

Ans: Inheritance is the OOP feature by which a new class (derived class) acquires the properties and behaviour of an existing class (base class), allowing code reusability and the creation of a hierarchical class structure.

(e) What is function overloading ?

Ans: Function overloading is a feature of C++ where two or more functions have the same name but differ in the number or type of parameters. The correct function is selected by the compiler at compile time based on the arguments passed — an example of compile-time polymorphism.

(f) What is constructor in OOP ?

Ans: A constructor is a special member function that has the same name as its class, has no return type, and is automatically called when an object of the class is created. It is mainly used to initialise the data members of the class.

(g) Give examples of non-linear data structure.

Ans: Examples of non-linear data structures: Tree and Graph — in these, data elements are not arranged sequentially and an element can connect to more than one other element.

(h) What is pointer ?

Ans: A pointer is a variable that stores the memory address of another variable, rather than a data value itself. It is declared using the '*' symbol, e.g., `int *p;`

(i) Write the full form of SQL.

Ans: The full form of SQL is Structured Query Language.

(j) What is WAN ?

Ans: WAN stands for Wide Area Network. It is a network that covers a very large geographical area, such as a country or across several countries, connecting multiple LANs and MANs together, e.g., the Internet.

(k) Write two examples of web browser.

Ans: Two examples of web browsers: Google Chrome and Mozilla Firefox. (Other examples: Microsoft Edge, Safari, Opera.)

(l) What is destructor in OOP ?

Ans: A destructor is a special member function that has the same name as its class preceded by a tilde (~) symbol, has no return type and no parameters, and is automatically called when an object goes out of scope or is deleted, in order to free/release the resources used by that object.

(m) What is relational algebra ?

Ans: Relational algebra is a procedural query language used in relational database management systems (RDBMS) that takes relations (tables) as input and produces a new relation as output, using operations such as SELECT, PROJECT, UNION, JOIN, and DIFFERENCE, to retrieve and manipulate data.

Q. No. 2 — Answer any ten ($2 \times 10 = 20$) [All parts solved]

(a) What is the difference between a class and an object ?

Ans: A class is a user-defined blueprint/template that defines the data members and member functions common to all objects of that type, but it does not occupy memory by itself. An object is an instance of a class, created using that blueprint, which occupies memory and holds actual values for the data members. In short: class is the design, object is the real entity built from that design.

(b) Define member function.

Ans: A member function is a function declared inside a class that operates on the data members of that class, defining the behaviour of the objects created from it, e.g., `getData()` and `putdata()`.

(c) What is the use of scope resolution operator (::) ?

Ans: The scope resolution operator (::) is used in C++ to: (1) define a member function of a class outside the class body (e.g., `void Class::function() { ... }`), and (2) access a global variable when a local variable of the same name exists in the current scope. It essentially links a member's declaration inside the class to its definition/implementation.

(d) What are the key features of OOP ?

Ans: The key features of Object-Oriented Programming (OOP) are: (1) Class and Object, (2) Encapsulation (binding data and functions together), (3) Data Abstraction (hiding implementation details), (4) Inheritance (acquiring properties of another class), (5) Polymorphism (one interface, many forms), and (6) Data Hiding (restricting access to data).

(e) What is the difference between private and protected access specifier ?

Ans: Private: members declared private are accessible only within the same class; they are not accessible in derived classes or outside the class. Protected: members declared protected are accessible within the same class AND within its derived (child) classes, but not from outside the class hierarchy. Thus, 'protected' is like 'private' but with an additional exception for inheritance.

(f) What is a stack ? Mention its two applications.

Ans: A stack is a linear data structure that follows the LIFO (Last In First Out) principle, where insertion and deletion both take place from the same end, called the 'top'. Two applications: (1) Managing function calls/recursion in a program, (2) Implementing the 'Undo' operation in text/word editors. (Also used in expression evaluation — infix to postfix conversion.)

(g) What is searching ? Mention any two searching techniques.

Ans: Searching is the process of finding the location of a particular element (key) within a given collection of data (such as an array or list). Two searching techniques: (1) Linear Search (Sequential Search), (2) Binary Search.

(h) Write the syntax of 'create table' SQL command.

Syntax:

```
CREATE TABLE table_name
```

```
(
    column1 datatype,
    column2 datatype,
    column3 datatype,
    ...
);
```

Example:

```
CREATE TABLE Student (Roll_No INT, Name VARCHAR(30), Marks INT);
```

(i) What is SOP in Boolean algebra ?

Ans: SOP stands for 'Sum of Products'. It is a form of expressing a Boolean expression as an OR (sum) of two or more AND (product) terms, e.g., $F = A.B + A'.C$.

(j) Define associative law and commutative law of Boolean algebra.

Ans: Associative law: It states that the grouping of variables in an AND or OR operation does not affect the result, i.e., $A + (B + C) = (A + B) + C$, and $A.(B.C) = (A.B).C$. Commutative law: It states that the order of variables in an AND or OR operation does not affect the result, i.e., $A + B = B + A$, and $A.B = B.A$.

(k) Define Internet.

Ans: The Internet is a global network of interconnected computer networks that uses the standard TCP/IP protocol to communicate and share data, resources and services (such as e-mail, web browsing, file transfer) among millions of devices worldwide.

(l) What is the use of twisted pair cable ?

Ans: A twisted pair cable is a type of guided/wired transmission medium consisting of pairs of insulated copper wires twisted together, used to transmit data/telephone signals over a network while reducing electromagnetic interference (crosstalk) between the wires. It is commonly used in LANs (e.g., Ethernet cables) and telephone lines because it is low-cost and easy to install.

(m) Define TCP/IP.

Ans: TCP/IP (Transmission Control Protocol/Internet Protocol) is the standard set (suite) of communication protocols used to connect devices on the Internet and other networks. TCP is responsible for reliable, ordered delivery of data between applications, while IP is responsible for addressing and routing data packets to their correct destination.

(n) What is FTP ?

Ans: FTP stands for File Transfer Protocol. It is a standard network protocol used to transfer (upload/download) files between a client and a server over a TCP/IP network such as the Internet.

(o) What is candidate key ?

Ans: A candidate key is a column, or a minimal set of columns, in a table that can uniquely identify each record/row in that table. A table may have more than one candidate key, and one of them is chosen to become the primary key; the rest are called alternate keys.

Q. No. 3 — Answer any nine ($3 \times 9 = 27$) [All parts solved]

(a) Explain the concept of polymorphism.

Ans: Polymorphism means 'many forms'. It is an OOP feature that allows the same function name, method, or operator to perform different tasks depending on the context/object involved — e.g., a function `area()` may calculate the area of a circle, square, or rectangle differently depending on which object calls it. It is of two types: (1) Compile-time polymorphism (achieved through function overloading and operator overloading), and (2) Run-time polymorphism (achieved through function overriding using virtual functions).

(b) Explain the concepts of base class and derived class.

Ans: Base class (parent class) is the existing/original class whose properties (data members) and behaviour (member functions) are inherited by another class. Derived class (child class) is the new class created using inheritance, which acquires the members of the base class and can also add its own new members or modify/override existing ones. This relationship enables code reusability, e.g.: `class Student : public Person` — here Person is the base class and Student is the derived class.

(c) Define queue. Mention two applications of queue.

Ans: A queue is a linear data structure that follows the FIFO (First In First Out) principle, where insertion (enqueue) takes place at the rear end and deletion (dequeue) takes place at the front end. Two applications: (1) CPU/Process scheduling in operating systems, (2) Managing requests in a printer queue (print spooling). (Also used in call-centre systems and handling requests on a shared resource.)

(d) Differentiate between insertion and selection sorting techniques.

Ans: Insertion sort: (1) Builds the sorted array one element at a time by picking each element and inserting it into its correct position among the already-sorted part. (2) It is efficient/fast for small or nearly-sorted data sets. (3) It requires shifting of elements to make space for insertion. Selection sort: (1) Repeatedly selects the minimum (or maximum) element from the unsorted part and places it at the beginning (or end) of the sorted part. (2) It always performs the same number of comparisons, regardless of the initial order of data. (3) It requires only swapping, not shifting of many elements.

(e) Give some examples of DDL and DML commands.

Ans: DDL (Data Definition Language) commands — used to define/modify the structure of database objects: CREATE, ALTER, DROP, TRUNCATE. DML (Data Manipulation Language) commands — used to manipulate/manage the data stored in tables: INSERT, UPDATE, DELETE, SELECT.

(f) Draw the truth tables for AND, OR and NOT gates.

AND gate ($Y = A.B$):

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR gate ($Y = A+B$):

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate ($Y = A'$):

A	Y
0	1
1	0

(g) Simplify the Boolean expression $(A + B)(A' + C)$ using Boolean algebra.

Solution:

$$\begin{aligned} & (A + B)(A' + C) \\ &= A.A' + A.C + B.A' + B.C && \text{[multiplying out]} \\ &= 0 + AC + A'B + BC && \text{[since } A.A' = 0\text{]} \\ &= AC + A'B + BC \end{aligned}$$

Now applying the consensus theorem (BC is redundant when AC and A'B are present):
 $= AC + A'B$

Therefore, $(A + B)(A' + C)$ simplifies to $AC + A'B$.

(h) Give examples of different data transfer rates.

Ans: Examples of data transfer rate units: (1) bps – bits per second, (2) Kbps – Kilobits per second (1,000 bps), (3) Mbps – Megabits per second (1,000 Kbps), (4) Gbps – Gigabits per second (1,000 Mbps). For example, a home broadband may offer 100 Mbps, while a fibre-optic backbone link may offer speeds in Gbps.

(i) Define HTML and XML.

Ans: HTML (HyperText Markup Language): It is a standard markup language used to create and design the structure/layout of web pages, using predefined tags (e.g., <html>, <body>, <p>) that are displayed by web browsers. XML (eXtensible Markup Language): It is a markup language designed to store and transport data in a structured format, using user-defined (custom) tags, and it focuses on describing data rather than displaying it.

(j) Explain the server-side scripting.

Ans: Server-side scripting refers to scripts (programs) that are executed on the web server before the resulting web page is sent to the client's browser. These scripts are used to interact with databases, process user input/forms, and generate dynamic web content based on requests. Examples of server-side scripting languages: PHP, ASP.NET, JSP, Python (Django/Flask), Node.js. Unlike client-side scripting (e.g., JavaScript), the source code of server-side scripts is not visible to the end user.

(k) Compare 1G with 2G technology.

Ans: 1G (First Generation): (1) Introduced in the early 1980s, it was purely analog technology. (2) It supported only voice calls, with poor voice quality and no data services, and no security (calls could be easily intercepted). 2G (Second Generation): (1) Introduced in the early 1990s, it was based on digital technology (e.g., GSM/CDMA). (2) It supported voice calls with better quality along with basic data services like SMS and low-speed internet (e.g., GPRS/EDGE), and offered better security through digital encryption.

(l) State the uses of different aggregate functions in SQL.

Ans: Aggregate functions perform a calculation on a set of values and return a single summary value. Common ones: (1) SUM() – calculates the total of a numeric column, (2) AVG() – calculates the average value, (3) COUNT() – counts the number of rows/records, (4) MAX() – returns the largest value in a column, (5) MIN() – returns the smallest value in a column. These are typically used along with GROUP BY to summarise grouped data, e.g., total salary department-wise.

(m) Explain the types of data file.

Ans: Data files can broadly be classified into: (1) Text file – stores data as readable ASCII/Unicode characters, e.g., .txt files; easy to read but takes more storage space. (2) Binary file – stores data in the form of 0s and 1s (machine-readable form), e.g., image, audio, executable files; more compact and faster to process but not human-readable directly. Additionally, based on organisation, files may also be classified as Sequential files (data accessed in order), Random/Direct access files (data accessed directly using a key/address), and Indexed sequential files (a combination using an index for faster access).

Q. No. 4 — Answer any three ($5 \times 3 = 15$) [All parts solved]

(a) Write a C++ program to create a matrix of order 3×3 .

```
#include <iostream>
using namespace std;
int main()
{
    int a[3][3], i, j;
    cout << "Enter elements of 3x3 matrix:" << endl;
    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
```

```

    {
        cout << "Element [" << i << "]"[" << j << "]: ";
        cin >> a[i][j];
    }
}
cout << "\nThe 3x3 Matrix is:\n";
for (i = 0; i < 3; i++)
{
    for (j = 0; j < 3; j++)
    {
        cout << a[i][j] << "\t";
    }
    cout << endl;
}
return 0;
}

```

(b) Simplify the Boolean expression using a 4-variable K-map: $F(W,X,Y,Z) = \Sigma(4,5,6,7,12,13,14,15)$. Obtain the product of sum expression after simplification.

Step 1 — Plot the K-map (WX as rows, YZ as columns; cells 4,5,6,7,12,13,14,15 = 1, all others = 0):

	YZ=00	YZ=01	YZ=11	YZ=10
WX=00 (0-3)	0	0	0	0
WX=01 (4-7)	1	1	1	1
WX=11 (12-15)	1	1	1	1
WX=10 (8-11)	0	0	0	0

Step 2 — Grouping (SOP form):

The two middle rows (WX = 01 and WX = 11) are entirely 1, forming one group of 8 cells. In this group, W varies (0 and 1), X is always 1, and Y, Z both vary — so only X remains constant.

Simplified SOP: $F(W,X,Y,Z) = X$

Step 3 — Converting to Product of Sums (POS):

Since $F = X$ in SOP form, and X alone is already a single literal, the equivalent POS form is simply the same single-variable expression (a single literal has the same form in both SOP and POS):

Product of Sum (POS) expression: $F(W,X,Y,Z) = X$

(This can also be verified from $F' = W'X' + WX' \rightarrow F' = X'$ (0-cells, WX=00 and WX=10 rows), so $F = (X')' = X$, confirming the result.)

(c) Write an algorithm to convert an infix expression to postfix using a stack. Explain with an example.

Algorithm (Infix to Postfix using Stack):

1. Create an empty stack and an empty output string.
2. Scan the infix expression from left to right, one symbol at a time.
3. If the symbol is an operand (letter/digit), add it directly to the output string.
4. If the symbol is an opening bracket '(', push it onto the stack.
5. If the symbol is a closing bracket ')':
Pop and add operators from the stack to the output until an opening bracket '(' is encountered; discard both brackets.
6. If the symbol is an operator (+, -, *, /, ^):
While the stack is not empty AND the top of stack has an operator of precedence $\geq$ the precedence of current operator:
Pop the operator from the stack and add it to the output.
Push the current operator onto the stack.
7. After the whole expression is scanned, pop all remaining operators from the stack and add them to the output string.

8. The output string now contains the postfix expression.

Example: Convert $A + B * C$ to postfix.

Symbol	Stack	Output
A	(empty)	A
+	+	A
B	+	AB
*	+ *	AB
C	+ *	ABC
end	(pop all)	ABC*+

Postfix expression = A B C * +

(d) Explain three different network topologies.

Ans: (1) Bus topology – All the devices in the network are connected to a single common central cable, called the bus/backbone. It is simple and inexpensive for small networks, but if the main cable fails, the entire network is affected. (2) Star topology – All devices are individually connected to a central device (hub/switch). It is easy to install, manage and troubleshoot, and failure of one node does not affect the others, but if the central hub fails, the whole network goes down. (3) Ring topology – Every device is connected to exactly two neighbouring devices, forming a closed loop, and data travels around the ring (often using token passing). It performs consistently under heavy traffic, but failure of a single device/cable can disrupt the whole ring unless a dual-ring/backup path is used.

(e) Write a C++ program to calculate the area of a rectangle using a class 'rectangle'.

```
#include <iostream>
using namespace std;
class rectangle
{
    private:
        float length, breadth;
    public:
        void getData()
        {
            cout << "Enter length: "; cin >> length;
            cout << "Enter breadth: "; cin >> breadth;
        }
        float area()
        {
            return length * breadth;
        }
};
int main()
{
    rectangle r;
    r.getData();
    cout << "Area of the rectangle = " << r.area() << endl;
    return 0;
}
```

Sample Output:

```
Enter length: 10
Enter breadth: 5
Area of the rectangle = 50
```

(f) Differentiate between Wi-Fi and WiMAX.

Ans: Wi-Fi (Wireless Fidelity): (1) Based on IEEE 802.11 standard. (2) Provides wireless network coverage over a short range, typically within a building/campus (a few hundred metres), suited for LAN/hotspot use. (3) Generally offers lower power consumption and is cheaper to deploy for small areas. WiMAX (Worldwide Interoperability for Microwave Access): (1) Based on IEEE 802.16 standard. (2) Provides wireless broadband coverage over a much larger area (several kilometres), suited for MAN/WAN-level connectivity. (3) Designed for higher data rates over longer distances, typically used by ISPs to provide last-mile broadband access.